Structural Efficiency in Artificial Syntax: A Quantitative Comparison of Human, LLM, and Stochastic Baselines

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Abstract

Large Language Models are often evaluated on semantic fluency, yet their structural alignment with human cognitive architecture remains poorly understood. This research quantifies the latent structural constraints of natural language through a comparative graph-theoretical analysis of human, LLM, and random dependency trees across ten languages. We extracted roughly 18,000 syntactic graphs and computed five topological metrics to map the divergence between biological and machine syntax. Statistical validation via Kolmogorov-Smirnov tests reveals model-specific patterns in replication of Dependency Distance Minimization constraints. Structural analysis reveals a narrowing of syntactic breadth in LLM outputs; these trees exhibit significantly lower recursive depth paired with rigidly standardized branching factors, suggesting a failure to replicate the natural variance of human syntax. This study also provides an insight into the "Machine Signature" in text and highlights a fundamental divergence in how artificial and biological systems navigate syntactic complexity. Finally, the existence of a bias due to English dominant training by analysing cross-linguistic data is also explored.

1. Introduction & Research Problem

Human language exhibits well-documented structural regularities in its syntactic organisation. Dependency grammars represent sentence structure as directed acyclic graphs where tokens are nodes and syntactic relations are directed edges. Prior work has shown that these graphs are far from random: they minimise dependency length, avoid crossings, and display characteristic branching patterns [1, 2].

Large language models (LLMs) are trained exclusively on human-produced text, yet it is unclear whether they reproduce these deep structural regularities or merely approximate surface statistics. This paper addresses this gap through a systematic empirical comparison across ten languages and two models.

1.1 Research Questions

1. How strong are the known structural constraints of natural language dependency trees, quantitatively, when measured against randomly generated baselines?
2. Do dependency structures produced by LLMs reproduce these constraints, or do they diverge from human language in measurable ways?
3. Is there evidence of an "English bottleneck/bias" when LLMs are made to produce sentences in different languages? Due to major training material on the web being available in English.

1.2 Hypothesis

The dependency structure of a sentence can be represented as a directed acyclic graph (DAG), where each token is a node and syntactic relations are directed edges. Prior research has established that these structures exhibit systematic constraints in natural language. We hypothesise:

- **H_1:** Human dependency trees will differ significantly from random baselines, confirming that natural language syntax is governed by systematic constraints.
- **H_2:** LLM-generated trees will largely resemble human language but may show measurable divergence in specific metrics: shallower tree depth, model-dependent dependency lengths, and different arity distributions.
- **H_3:** The random baseline acts as a null control, properties where both human and LLM output differ from the random baseline indicate learned constraints. Properties where only human language differs point to constraints the models have not learned.

1.3 Objectives

1. To isolate the 'Machine Signature' in syntax by statistically comparing the mean dependency length (MDL) and hierarchical depth of LLM outputs against human gold-standard trees across 10 typologically diverse languages.
2. Identify metrics on which LLM-generated syntax, per model (ChatGPT, Gemini) diverges from human syntax.

2. Methods

2.1 Data Collection

Natural language sentences are sourced from annotated dependency corpora in CoNLL-U format, specifically the Syntactic Universal Dependencies (SUD) treebanks. Two parallel LLM datasets are constructed by prompting each model to generate text matched in style and sentence length to the particulars of the respective natural language corpus:

- **ChatGPT** (OpenAI GPT-5.3)
- **Gemini** (Google Gemini 3 Flash)

2.2 Dependency Graph Construction

Each sentence is represented as a dependency graph where tokens are nodes and syntactic relations are directed edges. For SUD sentences, trees are extracted directly from gold-standard annotations. The raw LLM-generated sentence outputs were parsed and stored in the CoNLL-U format using an automatic syntactic dependency parser (Stanza) and then further analysis was conducted. Therefore it was tried that LLM sentence outputs should be of good quality so that eventual errors in parsing could be minimised. This would help ensure that the results are not contaminated.

2.3 Random Baseline Generation

For each sentence of N nodes, a random tree with N nodes is generated. This preserves sentence length while randomising syntactic structure, ensuring structurally meaningful comparisons. By matching node counts, graph density is strictly controlled and held constant across all comparisons.

2.4 Structural Metrics

To quantify the topological properties of the extracted dependency trees, we compute five graph-theoretical metrics across the four datasets (Human, ChatGPT, Gemini, Random):

- **Tree Arity:** Defined as the maximum out-degree of any node within the graph, $A(S) = \max_{v \in V} \deg_{out}(v)$, characterizing the branching intensity of the syntactic structure.
- **Tree Depth:** The maximum path length from the root node to any leaf node, serving as a proxy for the recursive complexity of the sentence.
- **Mean Dependency Length (MDL):** To account for varying sentence lengths, we normalize the total dependency length. This metric measures structural efficiency, where lower values indicate compliance with the principle of Dependency Distance Minimization (DDM):

$$\mathrm{MDL}(S) = \frac{1}{n} \sum_{i=1}^{n} |h_i - i| \tag{1}$$

  where h_i represents the head index of the i-th token.
- **Crossing Complexity:** A measure of non-projectivity, defined by the number of edge intersections when the tokens are ordered linearly. A crossing exists between edges (h_i, i) and (h_j, j) if:

$$\min(i, h_i) < \min(j, h_j) < \max(i, h_i) < \max(j, h_j) \tag{2}$$

2.5 Comparative Analysis

We perform three levels of pairwise distributional comparisons per language to isolate the "Machine Signature" in syntax:

- **Human vs. Random:** A validation step to confirm that natural language syntax adheres to systematic, non-random structural constraints.
- **LLM vs. Random:** A check to verify that LLM outputs are governed by learned syntactic rules rather than stochastic generation.
- **Human vs. LLM:** The primary investigative axis, identifying the specific mathematical divergence between biological and artificial syntactic production.

Full empirical distributions are analyzed using overlaid Kernel Density Estimations (KDEs) for arity, depth, and MDL, while structural variance is assessed via box plots of graph density. To statistically quantify these divergences, we apply the two-sample **Kolmogorov–Smirnov (KS) test**. The KS statistic $D_{n,m}$ measures the maximum distance between the cumulative distribution functions $F(x)$ of the two samples:

$$D_{n,m} = \sup_x |F_{\text{human}}(x) - F_{\text{LLM}}(x)| \tag{3}$$

A result is considered significant if the associated p-value < 0.05.

3. Implementation

3.1 Language Sample

Ten languages are selected to ensure typological diversity across word order, morphology, and syntactic structure (Table 1). This range spans SOV, SVO, and VSO languages, as well as analytic, agglutinative, and inflected morphological types, making the structural comparisons more generalisable and reducing the risk of language-family bias.

Table 1: Language sample and typological properties.

Language	Family	Order	Morph.
English	Indo-European	SVO	Analytic
Hindi	Indo-European	SOV	Fusional
German	Indo-European	SOV/SVO	Fusional
Arabic	Semitic	VSO	Fusional
Japanese	Japonic	SOV	Agglutinative
French	Indo-European	SVO	Fusional
Turkish	Turkic	SOV	Agglutinative
Mandarin	Sino-Tibetan	SVO	Analytic
Russian	Indo-European	free	Fusional
Finnish	Uralic	SVO	Agglutinative

3.2 Dataset Summary

Table 2 summarises the number of sentences collected per language and per source.
The SUD corpora available for different languages *(en_ewt, de_gsd etc.)* vary in the type of sentences they contain, like Wikipedia, News, Web-text. So, for a better analysis, the LLM was prompted to generate sentences in the same style as that available in the corresponding SUD bank.
At the same time, it was also instructed to create variations in the sentence structure and minimise duplicates. Sentences were generated in chunks instead of going for 500 at once to ensure good quality of outputs and avoid hallucination.

Example prompt for English EWT data (Which are typically from weblogs, newsgroups, emails, reviews, and Yahoo answers):

```
Generate exactly 50 sentences in English
    that resemble informal but
    grammatically complete web writing,
    the kind found in online news comments
    , blog posts, and factual Q&A forums.
    Topics should span everyday factual
```

```
    knowledge: science, geography, current
     events, technology, sports, and
    history.

Vary the syntactic structure deliberately
     across the 50 sentences:
- At least 10 sentences should contain a
    subordinate clause (e.g. "Although the
     river floods annually, the town has
    never relocated.")
- At least 10 should contain a relative
    clause (e.g. "The enzyme that breaks
    down lactose is pr oduced in the small
     intestine.")
- At least 10 should be compound
    sentences joined by coordinating
    conjunctions
- The remaining sentences can be simple
    declarative

Each sentence must be between 12 and 20
    words long. Do not use passive voice
    exclusively mix active and passive
    constructions. Do not use bullet
    points, numbered lists, headings, or
    dialogue. Output only the sentences,
    one per line, with absolutely no
    additional commentary, labels, or
    blank lines between sentences.
```

For each language, the prompt was designed to make responses replicate respective sources completely in the corresponding language's script with the help of Claude.
Along with this, a 'sanity check' prompt was also given to ensure quality output. Some times extra instructions had to be added due to repeated cases of duplicate responses being generated.
Particulars of the prompts and instructions given for each language can be found in the file

```
'README-prompt-instructions.md'
```

linked here (GitHub).

Table 2: Sentence counts per language and source. The KS test is robust to unequal sample sizes.

Lang.	Human	LLM	Random
English	*400*	*473*	*400*
Hindi	*400*	*493*	*400*
German	*400*	*500*	*400*
Arabic	*400*	*497*	*400*
Japanese	*400*	*497*	*400*
French	*400*	*496*	*400*
Turkish	*400*	*497*	*400*
Mandarin	*400*	*497*	*400*
Russian	*400*	*500*	*400*
Finnish	*400*	*495*	*400*

3.3 Experiment 1 – Human vs. Random

The initial phase of the analysis establishes a null hypothesis by comparing human annotated trees from the SUD treebanks against length-matched random baseline trees. This experiment serves to validate the methodology and quantify the degree to which natural language syntax differs from stochastic graph configurations.

3.3.1 Structural Distribution Analysis

We observe a profound divergence between human and random structures across all five topological metrics. The most significant divergence is found in the **Mean Dependency Length (MDL)**, where the human baseline exhibits a sharp peak at MDL ≈ 2.45, while the random baseline exhibits a flat, high-variance distribution centered at MDL ≈ 6.44.

- **Efficiency Constraints:** The significant rightward shift of the random baseline in MDL confirms that human syntax is uniquely governed by the principle of Dependency Distance Minimization (DDM).
- **Recursive Depth:** Interestingly, human trees show a **greater** mean depth (≈ 5.94) compared to the random baseline (≈ 4.52). This indicates that natural language is fundamentally more hierarchical and recursive than a stochastic arrangement of tokens, which tends toward flatter configurations.
- **Density Control:** As per the length-matching protocol, the graph density distributions for Human and Random datasets were mathematically identical.

3.3.2 Projectivity and Crossing Complexity

The crossing complexity metric reveals the most distinct non-random signature. Human language is predominantly projective, yielding a mean crossing count of approximately **0.43**. In contrast, the random baseline yields a mean of **72.02** crossings per graph. This order-of-magnitude difference highlights the strict linear constraints under which human syntax operates, which are entirely absent in the random null state.

3.3.3 Statistical Significance

To formally reject the null hypothesis, we applied the two-sample Kolmogorov–Smirnov (KS) test to the Human and Random distributions for each metric.

Table 3: KS-Test Results for Experiment 1 (Human vs. Random)

Metric	D-Statistic	p-value	Significant?
MDL	0.618	< 0.001	True
Tree Depth	0.242	< 0.001	True
Arity	0.051	< 0.001	True
Crossings	0.799	< 0.001	True

The resulting p-values for all languages and metrics were consistently below the $\alpha = 0.05$ threshold, effectively confirming that the SUD treebank data represents a highly optimized structural system distinct from random graph theory. This established ground truth allows for a meaningful comparison with LLM-generated syntax in the subsequent experiment.

3.4 Experiment 2 – Human vs. LLM

Experiment 2 evaluates whether artificial language models (Gemini and ChatGPT) internalize the latent structural constraints of natural language. By comparing these models against the human ground truth and a stochastic baseline, we isolate a distinct "Machine Signature" characterized by efficiency loss and structural homogenization.

3.4.1 Syntactic Efficiency and MDL variation

Analysis of the Mean Dependency Length (MDL) reveals divergent model performance, with Gemini producing syntactically "expensive" structures compared to humans, while ChatGPT achieves near-human efficiency.

- **Efficiency Divergence:** While humans optimize for low-cost dependencies (mean MDL ≈ 2.45), the machine outputs exhibit a slight rightward shift (Gemini MDL ≈ 2.86; ChatGPT MDL ≈ 2.46), here we see ChatGPT performing better than Gemini, and even getting really close to human results.
- **Cognitive Implications:** This suggests a model-specific pattern: ChatGPT has successfully internalized DDM constraints (MDL ≈ 2.46, nearly identical to humans at 2.45), while Gemini (MDL ≈ 2.86) appears to prioritize surface fluency over structural efficiency. This divergence indicates that the learning of DDM is not an inevitable outcome of language model training, but depends on architecture-specific or training-specific factors.

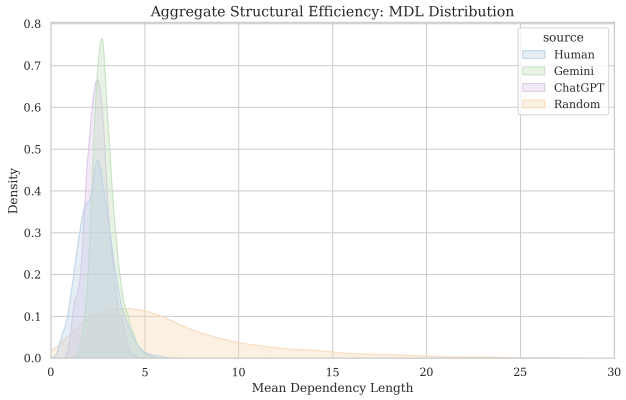

Figure 1: Empirical distribution of Mean Dependency Length (MDL) across ten languages.

3.4.2 Hierarchical Depth and "Shallow" Syntax

The distribution of recursive complexity (see Fig. 2) exposes a fundamental topological divergence.

- **Recursive Capping:** Human syntax is characterized by deep, recursive hierarchies (mean depth ≈ 5.94). In contrast, ChatGPT trees are significantly shallower (mean ≈ 4.61), near the stochastic baseline (mean ≈ 4.52) but Gemini gets a bit closer to humans (mean ≈ 5.62). Even then, the mean value might not be of much significance as the plot indicates clear differences in the distribution.
- **Syntactic Homogenization:** The LLM depth distributions exhibit spikiness at specific intervals (depths 4 and 5), indicating a reliance on standardized templates. This "Machine Signature" suggests that current architectures favor probabilistic safeness over the fluid complexity of human language.

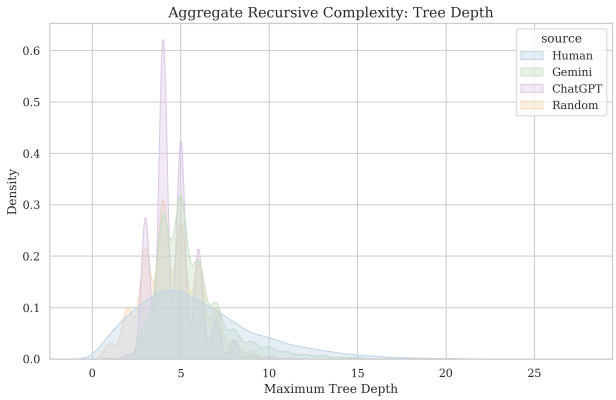

Figure 2: Comparative analysis of syntactic recursive depth. Human syntax (blue) shows a high-variance, deep recursive structure. Machine-generated syntax exhibits "Recursive Capping," with distributions clustered at shallower depths similar to the stochastic baseline.

3.4.3 Hyper-Projectivity

A counter-intuitive result is found in crossing complexity. LLMs produce significantly *fewer* crossings than humans (mean 0.17 vs. 0.43). We characterize this phenomenon as **Hyper-Projectivity.** An over-optimization of the projectivity constraint. By virtually eliminating non-projective arcs, LLMs fail to replicate the complex, non-linear stylistic variances that characterize natural human production.

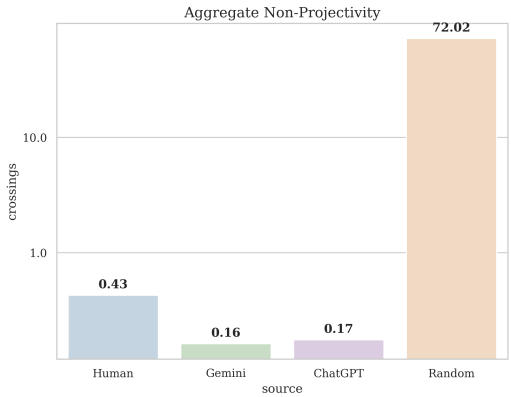

Figure 3: Aggregate Non-Projectivity (Mean Crossing Complexity). Note the logarithmic scale.

3.4.4 Statistical Summary

The Kolmogorov–Smirnov (KS) tests confirm that the structural divergence between humans and machines is statistically significant across all metrics ($p < 0.001$).

Table 4: KS-Test Results for Experiment 2 (Human vs. Individual LLMs)

Metric	$D_{ChatGPT}$	D_{Gemini}	p-values	Significant?
MDL	0.118	0.295	< 0.001	True
Tree Depth	0.283	0.192	< 0.001	True
Arity	0.188	0.357	< 0.001	True
Crossings	0.124	0.125	< 0.001	True

3.5 Experiment 3 – Cross-Linguistic Typological Analysis

The final experiment investigates the "English Bottleneck" hypothesis: the theory that LLM syntactic performance is optimized for high-resource, head-initial languages (like English) at the expense of typologically diverse structures. By analyzing ten languages across four language families, we identify where the machine signature diverges most sharply from human ground truth.

3.5.1 The English Bottleneck vs. Universal Efficiency

As visualized in the cross-linguistic distributions (Fig. 4), the efficiency is highly dependent on typological alignment.

- **High-Resource Alignment (English, French, Chinese):** In SVO languages like English (`en`) and French (`fr`), the peaks for ChatGPT and Gemini are sharply localized and track the Human baseline closely. Chinese (`zh`) exhibits similar strong localization, though it reveals a distinct model separation, with Gemini peaking at a visibly higher MDL than ChatGPT. Overall, this suggests that LLMs have successfully internalized the linear constraints of head-initial syntactic structures.
- **Rigid Templating in Non-SVO Languages (Hindi, Japanese, Turkish):** A dramatic shift occurs in head-final (SOV) and agglutinative languages. Crucially, rather than flattening toward the Random baseline, the LLM distributions in Japanese (`ja`) and Turkish (`tr`) form intensely sharp, narrow peaks that diverge significantly from the natural variance of the Human baseline. Hindi (`hi`) displays a similar divergence with a pronounced rightward shift toward higher MDLs. This indicates not a structural breakdown into randomness, but rather a reliance on highly constrained, machine-specific syntactic templates when operating outside the English-centric bottleneck.

3.5.2 Model Comparison: Gemini vs. ChatGPT

The facet grid reveals subtle but consistent differences in how the two models handle structural complexity:

- **Gemini Performance:** In languages like Russian (`ru`), Gemini maintains a notably tighter, spikier peak than ChatGPT. This suggests a rigid adherence to specific learned structural templates and a more conservative generation policy. However, this behavior is not universal; in German (`de`), the trend reverses, with ChatGPT exhibiting the sharper peak.
- **ChatGPT Performance:** In structurally divergent languages where the machine signature separates from the human baseline, ChatGPT tends to generate slightly more efficient dependencies. Specifically in Hindi and Turkish, Gemini shifts further to the right on the MDL spectrum, meaning ChatGPT maintains a lower mean dependency length. This makes ChatGPT comparatively closer to the human baseline in these SOV and agglutinative languages than its counterpart.

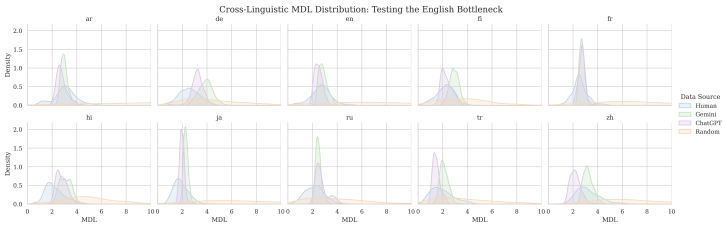

Figure 4: Cross-linguistic comparison of Mean Dependency Length (MDL). *Enlarged figure in appendix.*

4. Discussion of Results

The results of this study reveal a distinct *Machine Signature* in dependency syntax that departs significantly from both human production and stochastic noise. Our analysis across ten languages uncovers three primary structural divergences.

The Efficiency Divergence. As demonstrated in Experiment 2, the two LLMs show divergent performance in minimizing dependency distance compared to natural language. While the Human cohort maintains a mean MDL of ≈ 2.45, ChatGPT achieves near-human performance at ≈ 2.46, whereas Gemini shows a notable rightward shift to ≈ 2.86. The KS-test ($D_ChatGPT = 0.118, D_Gemini = 0.295, p < 0.001$) confirms statistical divergence for both models, though ChatGPT's efficiency loss is minimal.

Recursive Capping and Homogenization. The most statistically significant divergence observed was in Maximum Tree Depth ($D_ChatGPT = 0.283$). Human syntax is characterized by deep, high-variance recursive structures (mean ≈ 5.94). In contrast, ChatGPT trees were significantly shallower (mean ≈ 4.61), hugging the Random baseline (mean ≈ 4.52). The spikiness of the LLM depth distributions (see Fig. 2) suggests that models rely on a narrow set of standardized structural templates, a phenomenon we term *syntactic homogenization*.

Hyper-Projectivity. Counter-intuitively, LLMs produce significantly cleaner trees than humans in terms of linear crossings. With a mean crossing count of 0.17 compared to the human 0.43, LLMs appear to have over-learned the constraint of projectivity. This *Hyper-Projectivity* suggests that models avoid the complex, non-linear stylistic variances that human speakers use for emphasis or topicalization, favoring a "safe" but structurally sterile linearization.

The English Bottleneck. Experiment 3 explores the typological limitations of these models. When processing head-final (Hindi, Japanese) or agglutinative (Turkish) languages, the models default to highly rigid, non-human syntactic templates. While performance in high-resource SVO languages like English (`en`) and French (`fr`) closely mirrors natural human variance. The hyper localized, sharp MDL peaks observed in typologically diverse languages reveal a stark departure from the human baseline. This confirms that current LLMs do not simply fail into randomness outside the English centric bottleneck; instead, they over optimize, relying on constrained, machine specific generation policies that fail to capture the natural structural flexibility inherent in global linguistic diversity.

Model-specific Performance Differences. Model-Specific Performance. A striking finding is the substantial difference between ChatGPT and Gemini in structural efficiency. ChatGPT's mean MDL of 2.46 is virtually indistinguishable from the human baseline of 2.45, suggesting near-complete internalization of the DDM constraint. Even Gemini's worse mean MDL of 2.86 is still dramatically more efficient than the random baseline (6.44). The larger KS D-statistic for Gemini (0.295) compared to ChatGPT (0.118) confirms this model-level divergence. This suggests that the acquisition of structural constraints is not uniform across transformer architectures and may depend on training methodology, model size, or architectural choices in attention mechanisms.

Limitations. Human sentences use gold-standard annotations while LLM text relies on the automated Stanza parser. Hence parser errors could inflate the structural irregularities observed in the metrics of the LLM outputs.

5. Theoretical Discussion

The topological divergences identified in this study carry significant implications for our understanding of both biological language evolution and artificial language processing. By quantifying where machines successfully mimic human syntax and where they fundamentally diverge, we can better isolate the specific cognitive pressures that shape natural language.

5.1 Cognitive Economy and Distributional Learning

The fact that human dependency trees consistently minimize dependency length and crossing complexity relative to random baselines across all ten languages reinforces the hypothesis that communicative efficiency pressures, shape syntactic structure over evolutionary and developmental timescales [1]. The human baseline consistently occupies the most efficient region of the topological landscape (a leftward peak in MDL distributions), strongly supporting the claim that natural language is a highly optimized system constrained by finite working memory resources. Furthermore, this universal efficiency across typologically diverse language families suggests that the cognitive pressure to minimize working memory load acts as a primary evolutionary driver, actively selecting for syntactic structures that optimize information flow during real-time communication.

Furthermore, the degree to which syntax departs from a random structure reflects the strict constraints imposed by incremental parsing. Crucially, the areas where LLMs successfully match human topological baselines demonstrates that certain structural efficiency constraints can emerge organically through purely distributional learning, without embodied cognitive pressures.

5.2 The Embodied vs. Static Processing Gap

Despite mimicking surface-level efficiency, the distinct "Machine Signature" observed in this study, specifically *Recursive Capping* and *Hyper-Projectivity* suggests a fundamental difference in how artificial and biological systems process complexity. Humans produce complex syntax under the intense real-time pressures of spoken production, relying on deep recursion and non-projectivity to efficiently manage information flow and topicalization. In contrast, LLMs are trained on static text corpora. They do not experience the physiological rate processes or the embodied bottlenecks of real-time communication. Consequently, they default to structurally simpler, highly predictable configurations. The consistency of this homogenization across both Gemini and ChatGPT strongly implies that these are not model-specific bugs, but rather inherent architectural limitations of the standard transformer objective, which heavily prioritizes local token probability over global hierarchical complexity.

5.3 The Illusion of a Universal Artificial Grammar

Finally, this structural rigidity exposes a severe typological bias within current foundation models. The differences in LLM data observed in the properties of languages that are structurally different from English suggests that the universal grammar internalized by these models is heavily biased toward Western linguistic topologies. This "English Bottleneck" is likely the direct result of a lack of diverse training data, as foundation models are predominantly trained on English-centric web corpora. Until models are exposed to a structurally balanced typological landscape, this bias poses a fundamental challenge to the development of truly cross-linguistic syntactic agents and directly challenges the cognitive assumption that current foundation models process syntax through a universal, language-agnostic mechanism.

6. Conclusion

This study quantitatively compared human dependency syntax across ten typologically diverse languages against LLM outputs and stochastic baselines. While LLMs successfully mimic surface-level fluency, they fail to replicate the underlying topological signatures of natural language. Instead, they exhibit a distinct "Machine Signature" characterized by shallower trees and rigid linearity. These empirical findings demonstrate that current transformer architectures have not fully internalized the cognitive efficiency-complexity trade-offs inherent to biological syntax. To bridge this gap and overcome the severe "English Bottleneck" identified in this study, future research must determine whether typologically-aware training or fundamental architectural shifts in transformer mechanism are required to develop truly cross-linguistic syntactic agents.

A. Python Code

All the code and the LLM generated sentence dataset is available at:
https://github.com/saatviksingh77/cgs410-project

Some snippets are attached below:

```python
######## CONSTRUCTION OF GRAPHS USING networkx ########

import networkx as nx
def get_metrics_safe(sentence_nodes):
    G = nx.DiGraph()
    edges = []
    dep_length = 0
    crossings = 0

    nodes_indices = [int(token['id']) for token in sentence_nodes if isinstance(token['id'], int)]
    n_nodes = len(nodes_indices)

    if n_nodes > 80 or n_nodes < 2:
        return {'arity': 0, 'depth': 0, 'density': 0, 'dep_length': 0, 'crossings': 0, 'n_tokens': n_nodes}

    for token in sentence_nodes:
        # SAFETY CHECK: Skip if ID is not an int OR if head/id is missing (None)
        if not isinstance(token['id'], int) or token['id'] is None or token['head'] is None:
            continue

        child, head = int(token['id']), int(token['head'])

        if head != 0:
            G.add_edge(head, child)
            edges.append((head, child))
            dep_length += abs(head - child)

    # Tree Arity
    arity = max([d for n, d in G.out_degree()] + [0])

    # Tree Depth
    try:
        roots = [n for n, d in G.in_degree() if d == 0]
        depth = max([max(nx.shortest_path_length(G, source=r).values()) for r in roots] + [0])
    except:
        depth = 0

    # Density and MDL
    density = len(edges) / (n_nodes * (n_nodes - 1)) if n_nodes > 1 else 0
    norm_dep_length = dep_length / n_nodes

    # Crossings
    for i in range(len(edges)):
        for j in range(i + 1, len(edges)):
            e1, e2 = sorted(edges[i]), sorted(edges[j])
            if e1[0] < e2[0] < e1[1] < e2[1] or e2[0] < e1[0] < e2[1] < e1[1]:
                crossings += 1

    return {
        'arity': arity, 'depth': depth, 'density': density,
        'dep_length': norm_dep_length, 'crossings': crossings, 'n_tokens': n_nodes
    }
```

```python
######## GENERATING RANDOM BASELINE TREES ########

import random
def generate_random_tree_metrics(n_nodes):
    if n_nodes < 2 or n_nodes > 80:
        return {'arity': 0, 'depth': 0, 'density': 0, 'dep_length': 0, 'crossings': 0, 'n_tokens': n_nodes}

    nodes = list(range(1, n_nodes + 1))
    random.shuffle(nodes)
    tree_nodes, remaining_nodes = [nodes[0]], nodes[1:]
    simulated_tokens = [{'id': nodes[0], 'head': 0}]

    while remaining_nodes:
        parent = random.choice(tree_nodes)
        child = remaining_nodes.pop()
        simulated_tokens.append({'id': child, 'head': parent})
        tree_nodes.append(child)

    return get_metrics_safe(simulated_tokens)
```

```python
### significance testing (KS test)
#### across all languages and both metrics, comparing human vs llm

metrics = {'dep_length': 'MDL', 'depth': 'Depth', 'arity': 'Arity', 'crossings': 'Crossings'}
target_models = ['Gemini', 'ChatGPT']

print("\n--- AGGREGATE KS-TESTS (HUMAN VS. MODELS) ---")
for model in target_models:
    print(f"\nComparison: Human vs {model}")
    for col, label in metrics.items():
        h_val = df_clean[df_clean['source'] == 'Human'][col]
        m_val = df_clean[df_clean['source'] == model][col]
        d_stat, p_val = ks_2samp(h_val, m_val)
        print(f"{label:<10} | D-Stat: {d_stat:.3f} | p: {p_val:.2e}")
```

```python
###### AGGREGATE KS-TESTS (HUMAN VS. MODELS) ######
metrics = {'dep_length': 'MDL', 'depth': 'Depth', 'arity': 'Arity', 'crossings': 'Crossings'}
target_models = ['Gemini', 'ChatGPT']

print("\n--- AGGREGATE KS-TESTS (HUMAN VS. MODELS) ---")
for model in target_models:
    print(f"\nComparison: Human vs {model}")
    for col, label in metrics.items():
        h_val = df_clean[df_clean['source'] == 'Human'][col]
        m_val = df_clean[df_clean['source'] == model][col]
        d_stat, p_val = ks_2samp(h_val, m_val)
        print(f"{label:<10} | D-Stat: {d_stat:.3f} | p: {p_val:.2e}")
```

```python
# Aggregate means across all 13,000+ sentences
agg_results = df_clean.groupby('source')[['dep_length', 'depth', 'crossings']].mean()

print("\n--- AGGREGATE RESULTS (ALL LANGUAGES) ---")
print(agg_results)
```

B. Figures

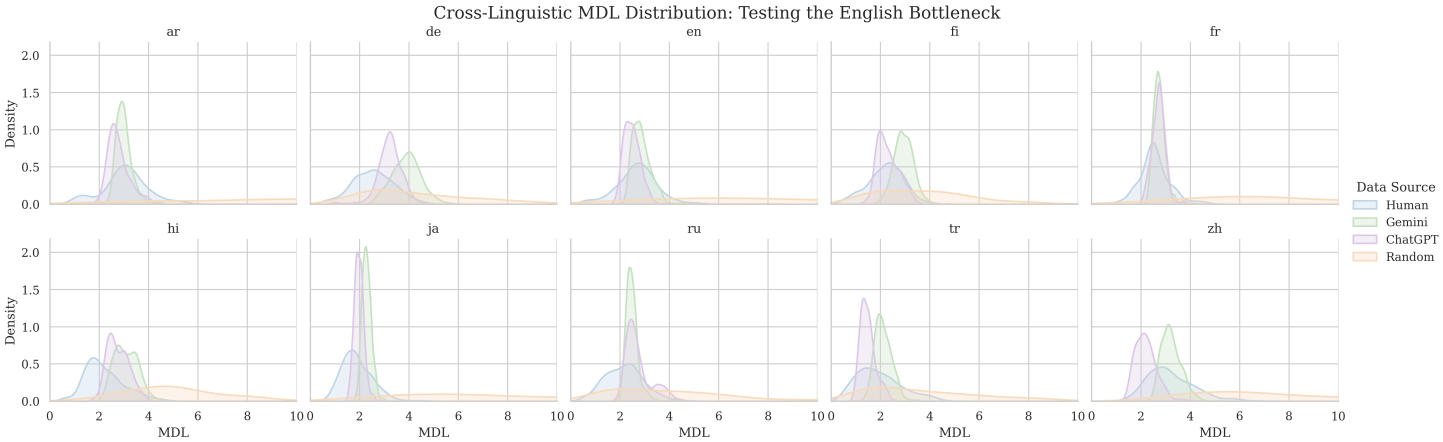

Figure 5: Cross-linguistic comparison of Mean Dependency Length (MDL) across 10 languages and 2 models.

References

[1] Richard Futrell, Kyle Mahowald, and Edward Gibson. Large-scale evidence of dependency length minimization in 37 languages. *Proceedings of the National Academy of Sciences*, 2015.

[2] Haitao Liu. Dependency distance as a metric of grammatical complexity. *Journal of Quantitative Linguistics*, 2008.